

Fundamentals of Fog and Edge Computing

BCSE313L

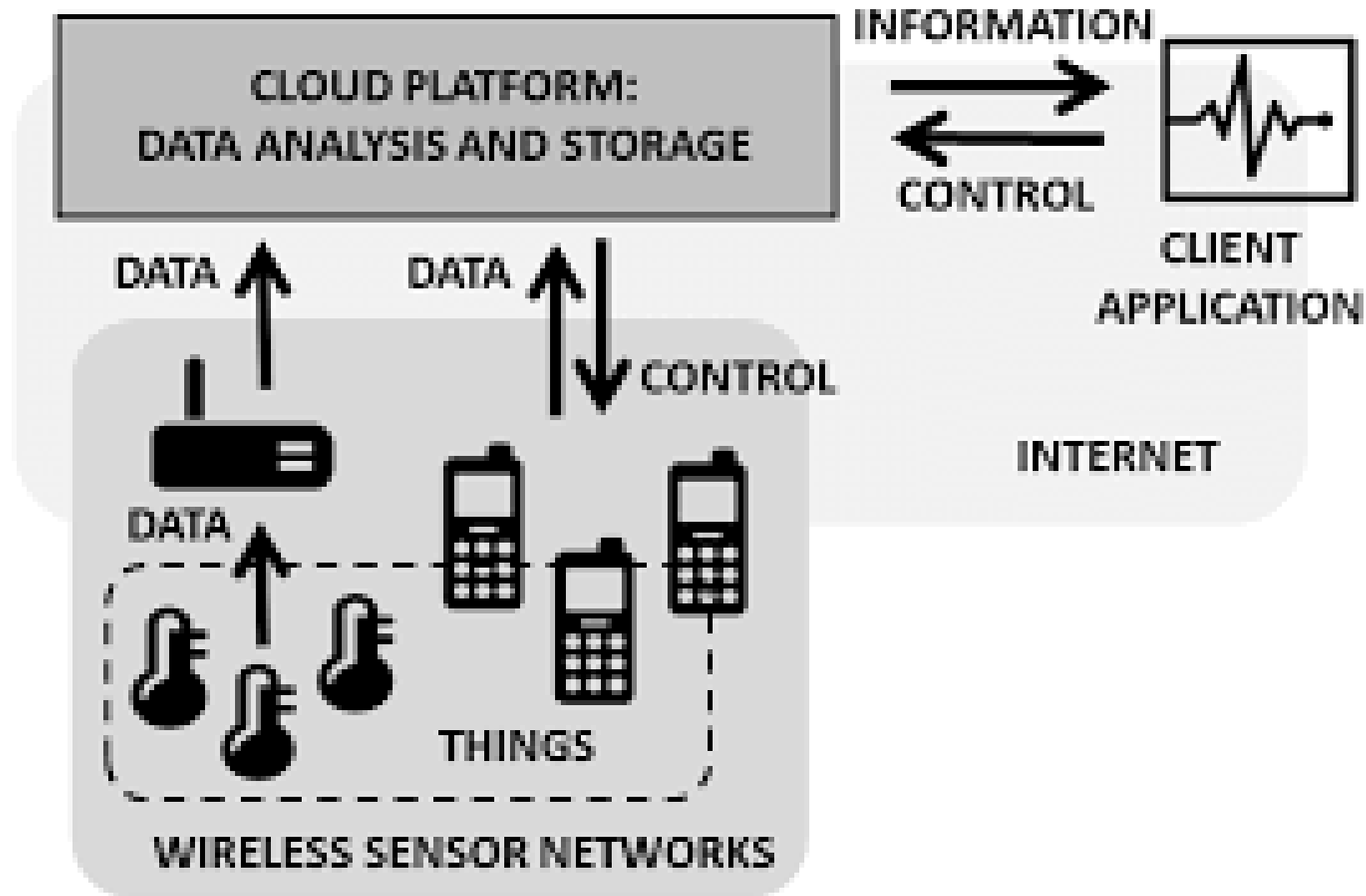
WINTER 2024 - 2025

**Module 1 - Internet of Things (IoT) and New Computing
Paradigms**

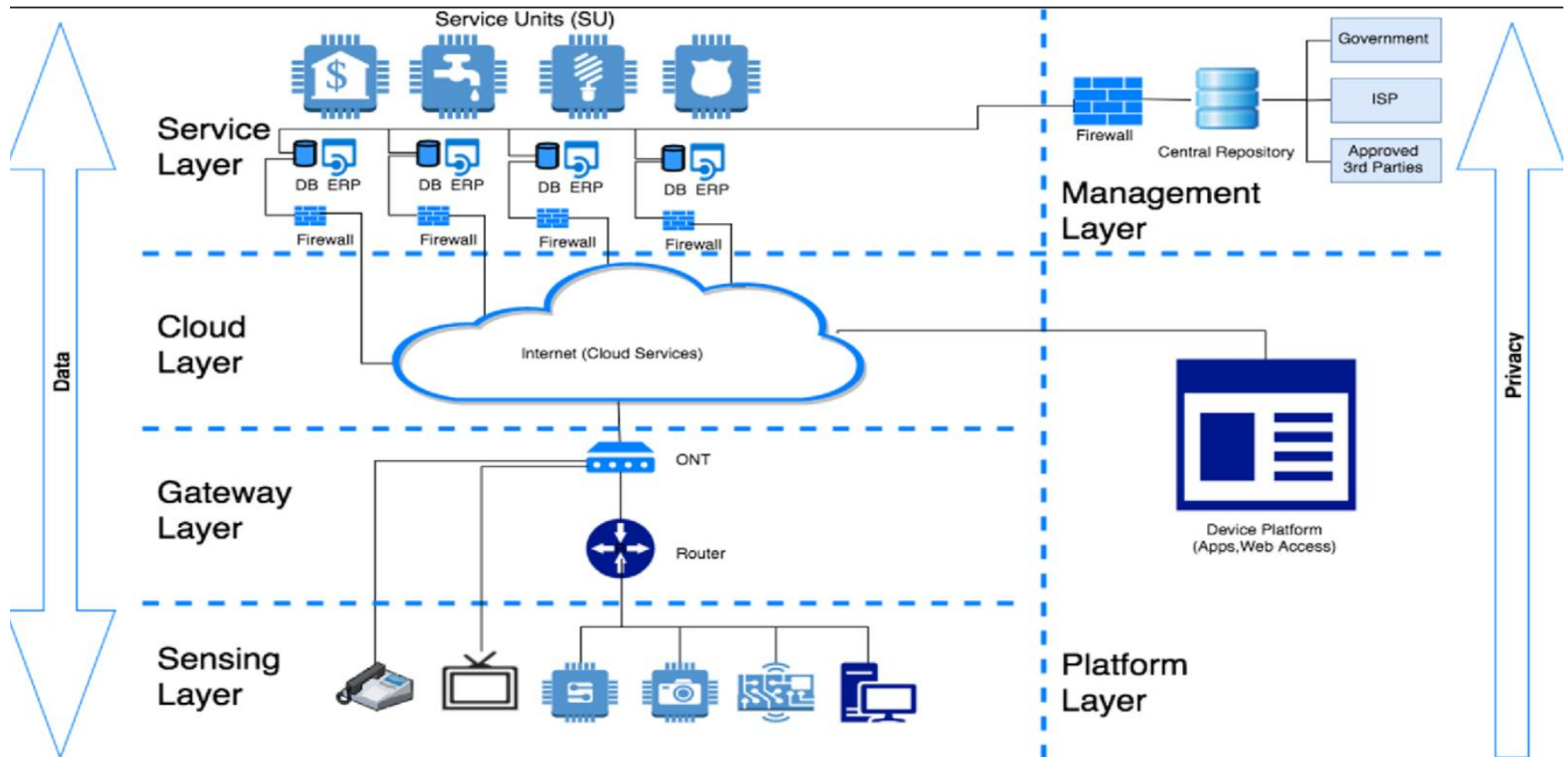
Introducing Internet of Things (IoT)

- **IoT as a Network of Physical Objects:** The IoT connects physical devices like appliances, vehicles, and sensors to the internet.
- **Enhanced Efficiency:** IoT improves the efficiency of various applications such as logistics, manufacturing, and agriculture.
- **Cloud-Centric IoT (CIoT) Architecture:** IoT systems often follow a cloud-centric architecture where physical objects are represented as web resources managed by cloud servers.

CIoT – Traditional Approach



CIoT – Traditional Approach



Introducing Internet of Things (IoT)

➤ **Key Components of IoT Systems:**

- **Front-end Devices:** Sensors, actuators, and readers interact with physical objects.
- **Gateways:** Devices like routers and modems connect front-end devices to the internet.

Introducing Internet of Things (IoT)

➤ **Core Technologies of IoT:**

- **Embedded Systems:** Provide intelligence to front-end devices.
- **Middleware:** Connects heterogeneous devices to the cloud.
- **Cloud Services:** Offer storage, processing, and management capabilities.

Introduction - Limitation of CIoT - BLURS

B

- Bandwidth – (Data Generated)

L

- Latency – (Delay or Time between)

U

- Uninterrupted Connectivity

R

- Resource Constraints (Compute Power)

S

- Security (Attacks on edge)

Introduction - Limitation of CIoT - BLURS

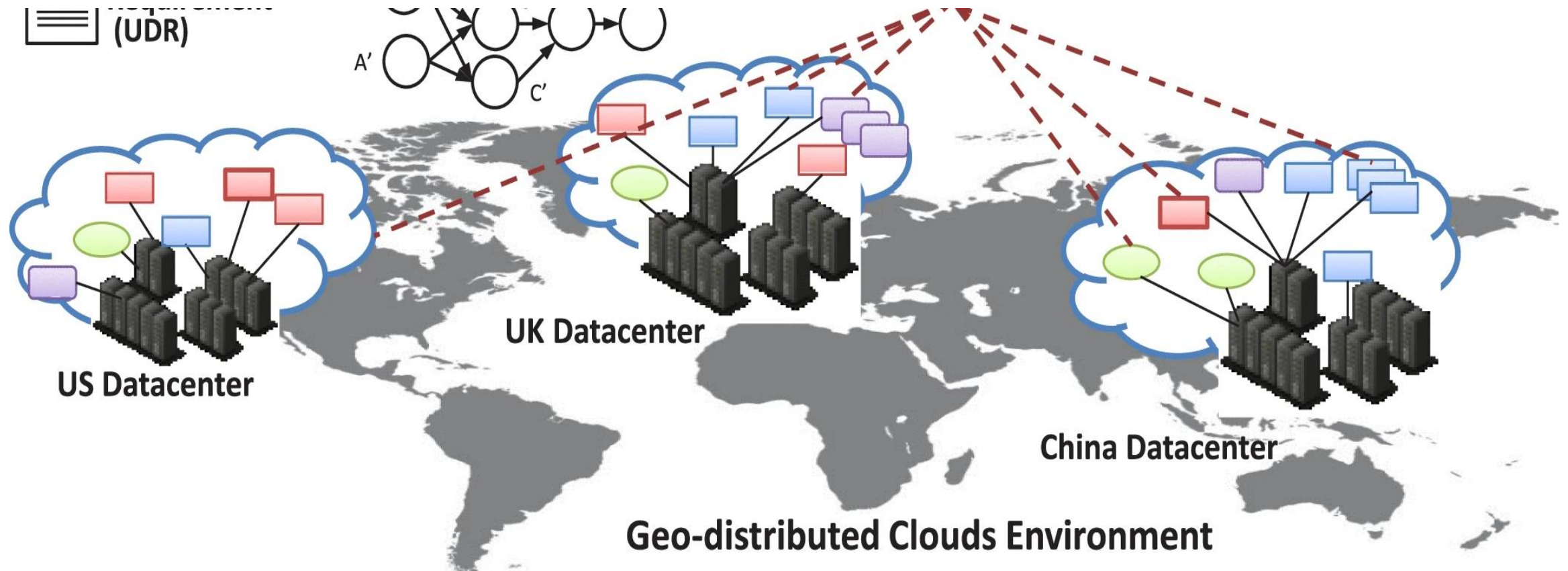
- **Bandwidth:** High-frequency data generation exceeds cloud bandwidth.
 - Connected cars and self-driving vehicles generate massive amounts of data.
- **Latency:** Cloud-based processing can introduce significant delays.
 - Industrial control systems, autonomous vehicles, and real-time applications require low latency.
- **Uninterrupted Connectivity:** Distance and network instability can disrupt service.
 - Network outages can disrupt IoT services, especially for remote devices.

Introduction - Limitation of CIoT

- **Resource Constraints:** Front-end devices may lack processing power and energy.
 - Battery-powered devices may struggle with continuous data transmission.
- **Security:** Remote devices are vulnerable to attacks.
 - Edge devices are susceptible to attacks and data manipulation.

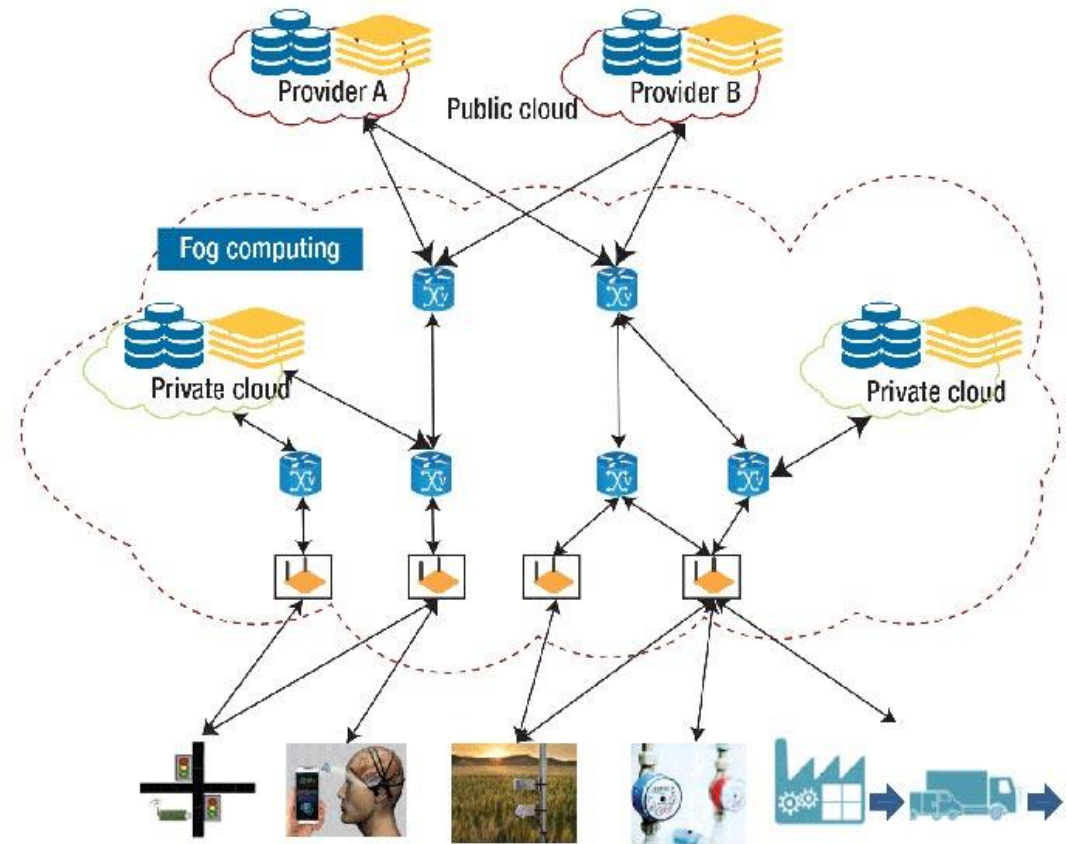
Introduction - Emerging Paradigms: Overcome Limitations of CIoT

- **Geo-Distributed Cloud Computing:** Distributes processing to data centers closer to the edge.



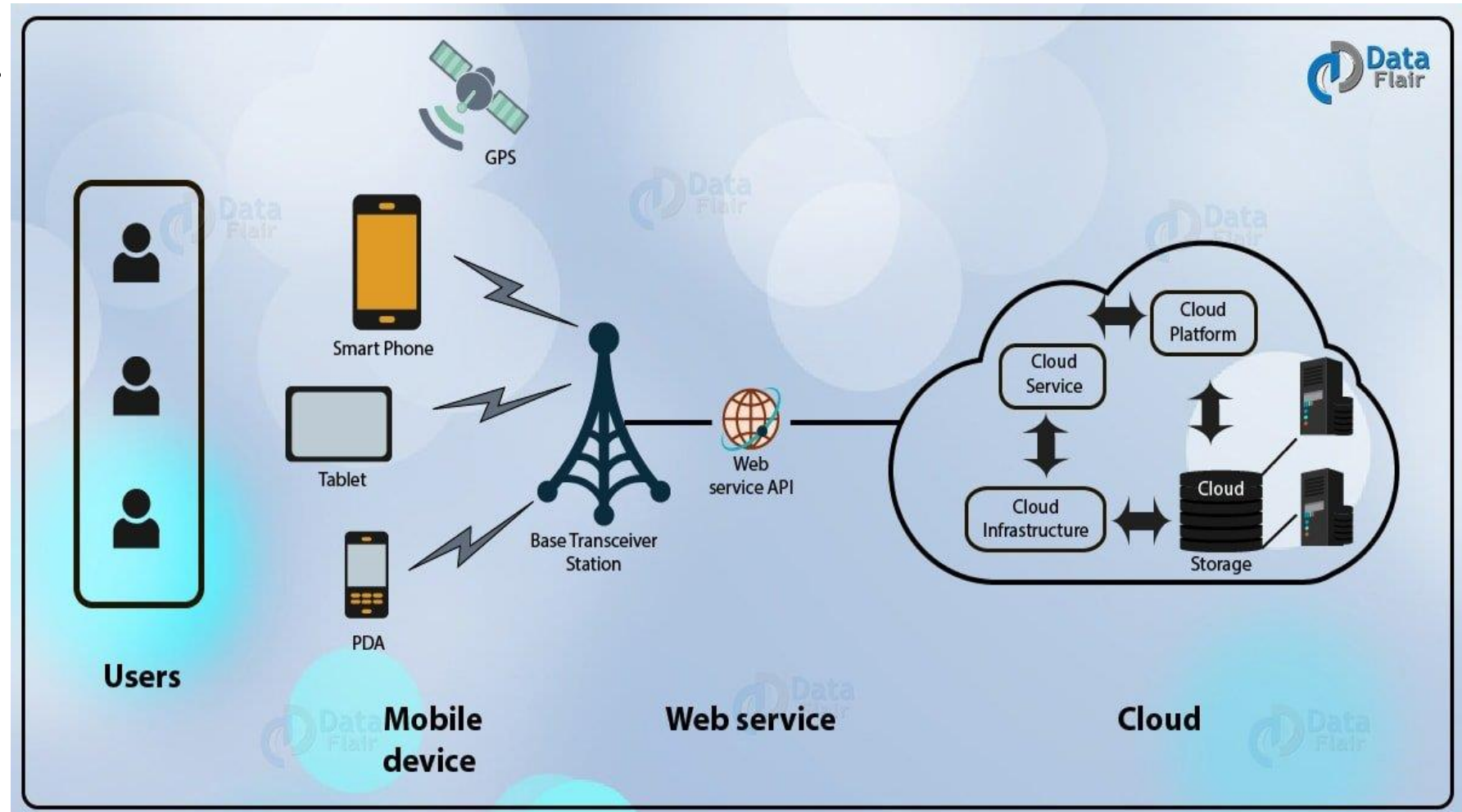
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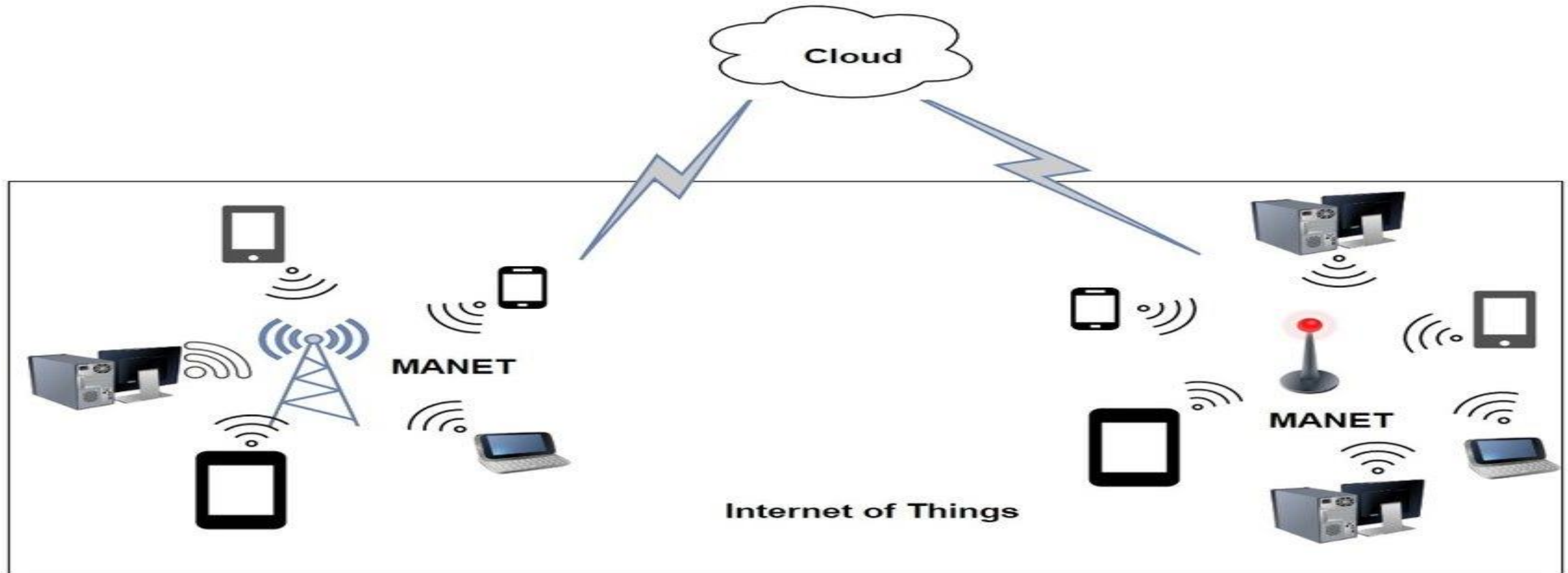
Introduction - Emerging Paradigms: Overcome Limitations of CIoT

➤ **Mobile Cloud Computing:** Leverages local wireless access points for cloud resources.



Introduction - Emerging Paradigms: Overcome Limitations of CIoT

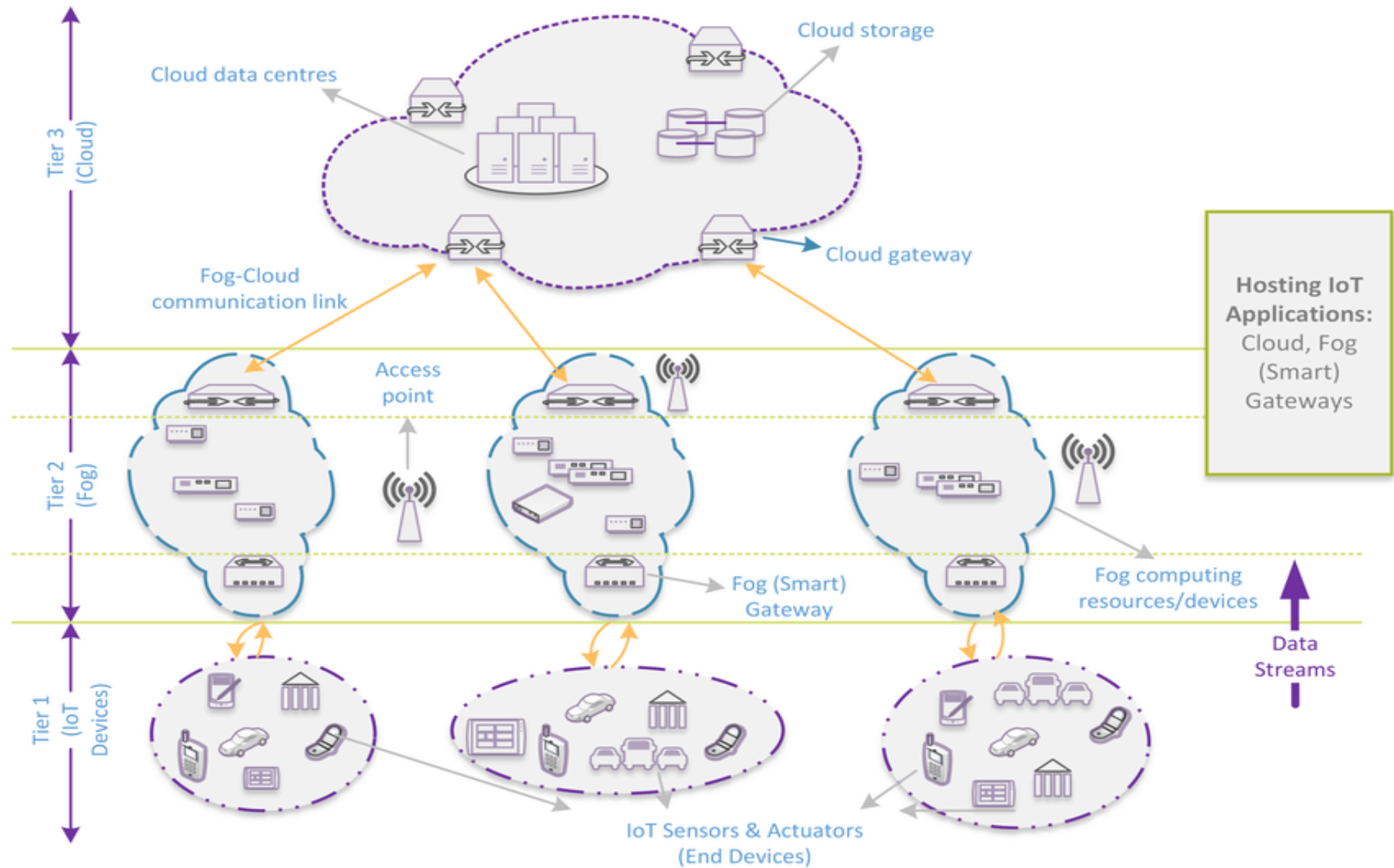
- **MANET-Based Cloud Computing:** Utilizes mobile devices to form ad-hoc networks for cloud services.



Introduction - Emerging Paradigms: Overcome Limitations of CIoT

➤ Fog Computing:

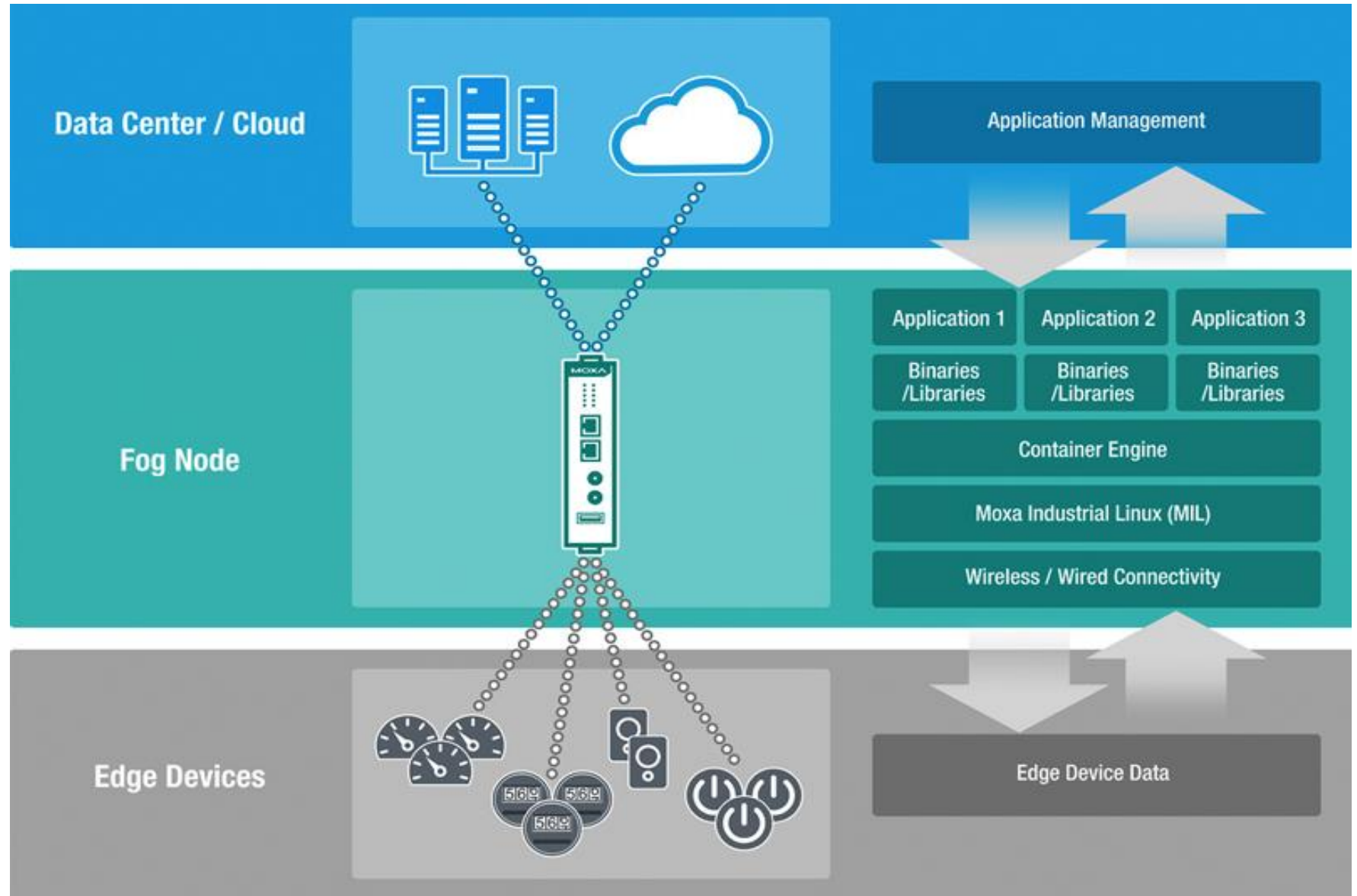
A hierarchical approach that distributes processing to devices at the edge of the network.



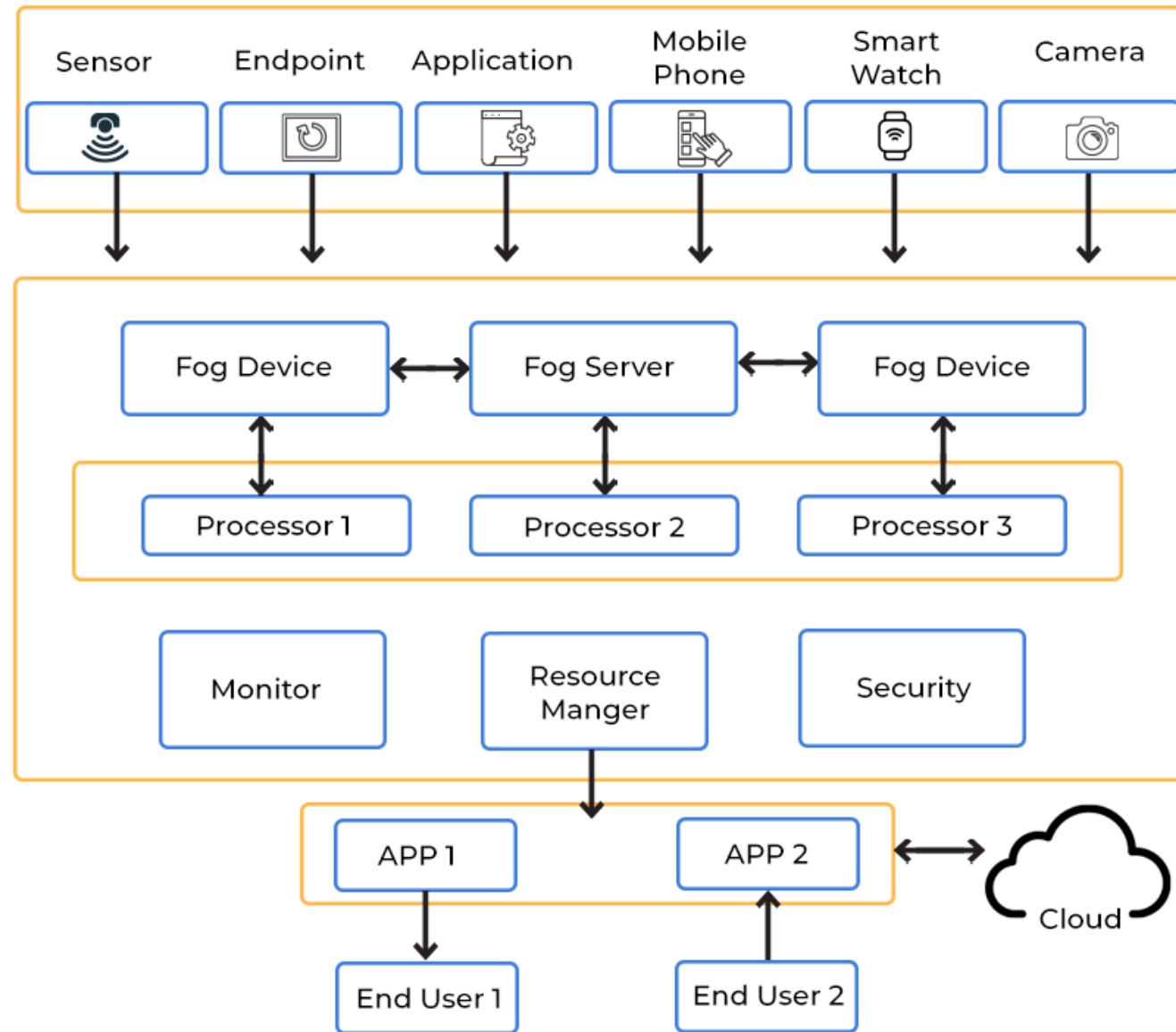
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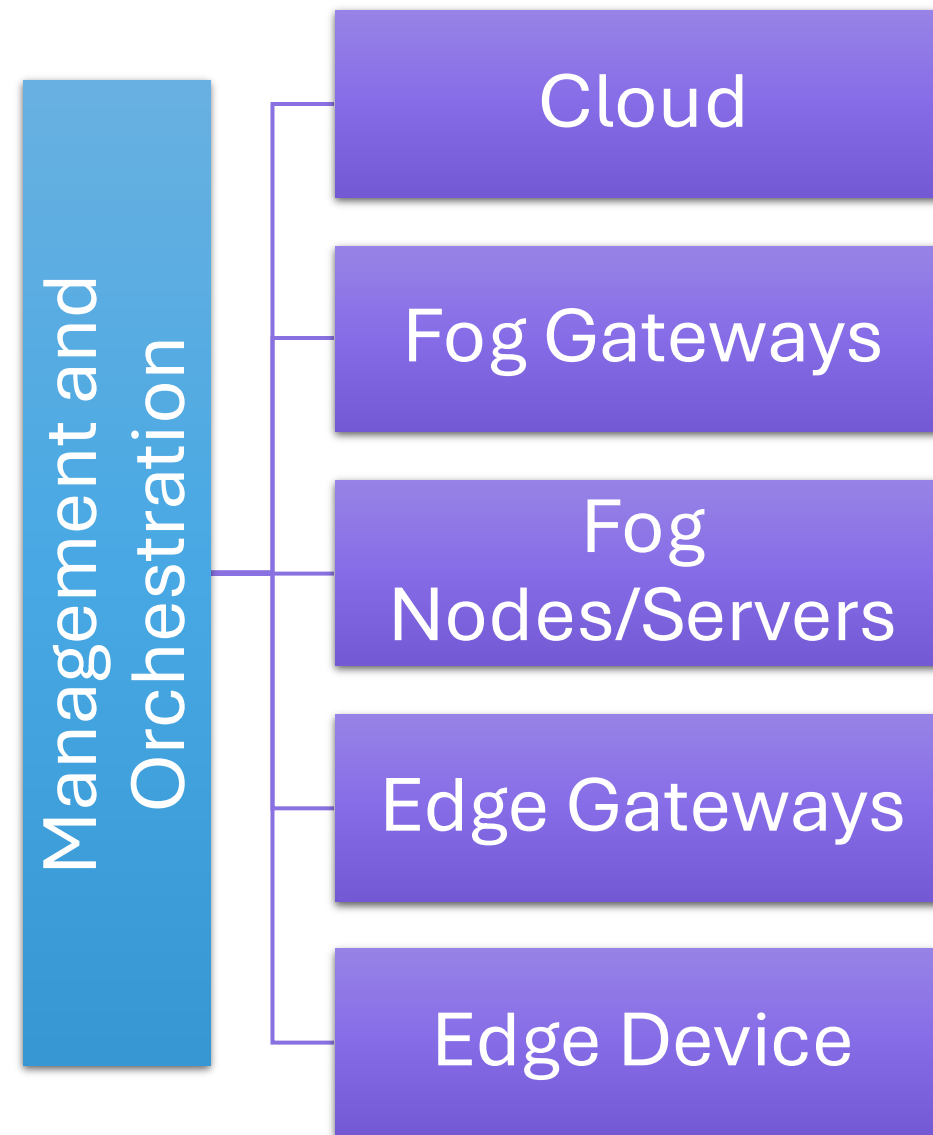
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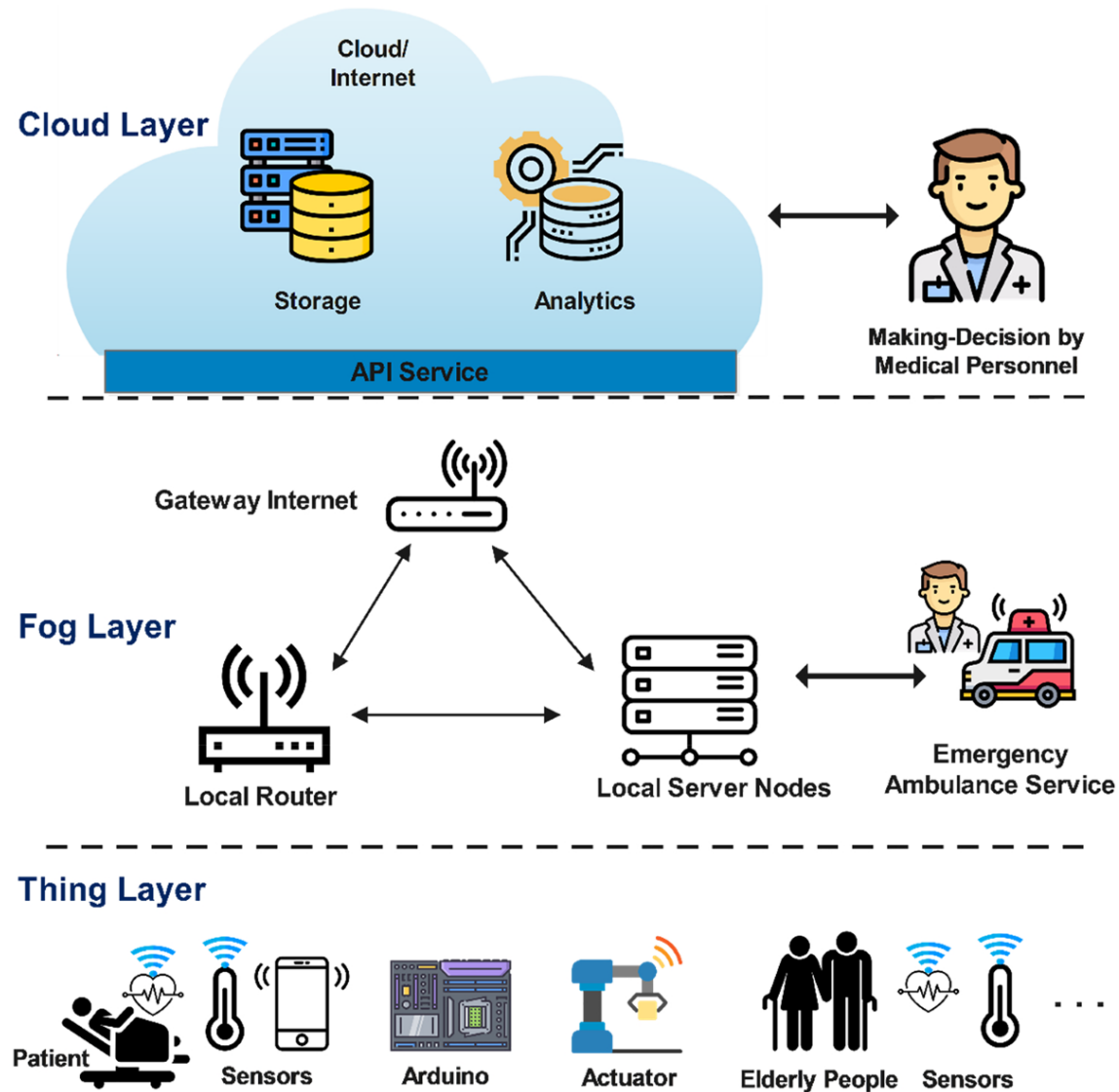
Introduction – Fog Computing - Architecture



Introduction – Fog Computing - Architecture



Introduction – Fog Computing - Architecture



Introduction – Fog Computing - Architecture

- **Hierarchical Model:** Involves multiple layers of devices, from edge devices to fog nodes and cloud servers.
- **Scalability:** Supports horizontal scaling to accommodate increasing workloads.
- **Flexibility:** Can adapt to different IoT scenarios and requirements.

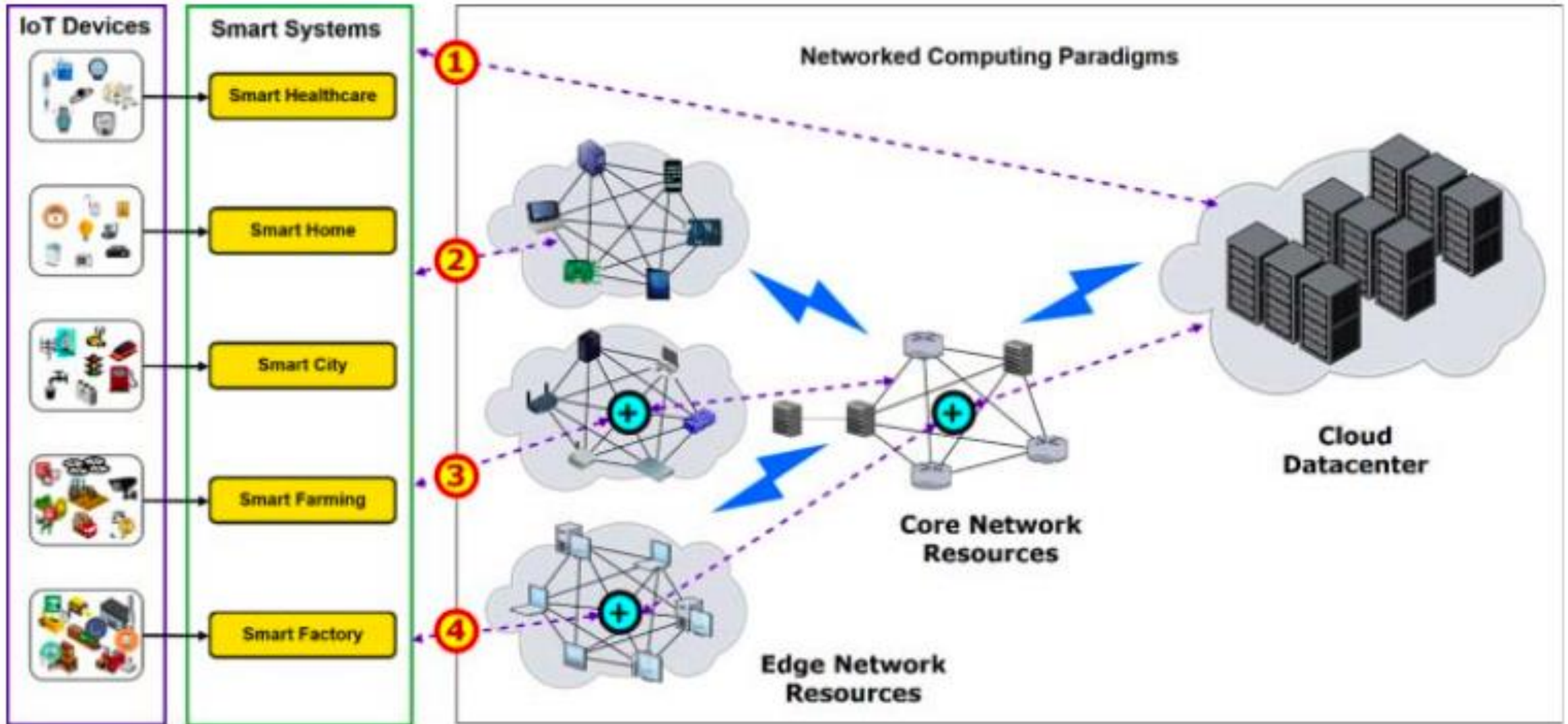
Introduction - Emerging Paradigms: Overcome Limitations of CIoT

- **Geo-Distributed Cloud Computing:** Distributes processing to data centers closer to the edge.
- **Mobile Cloud Computing:** Leverages local wireless access points for cloud resources.
- **MANET-Based Cloud Computing:** Utilizes mobile devices to form ad-hoc networks for cloud services.
- **Fog Computing:** A hierarchical approach that distributes processing to devices at the edge of the network.

Introduction – Fog Computing - Benefits

- **Reduced Latency:** Processing data closer to the source reduces response time.
- **Improved Bandwidth Utilization:** Offloads data processing from the cloud.
- **Enhanced Security:** Sensitive data can be processed locally, reducing security risks.
- **Increased Reliability:** Distributed architecture improves system resilience.

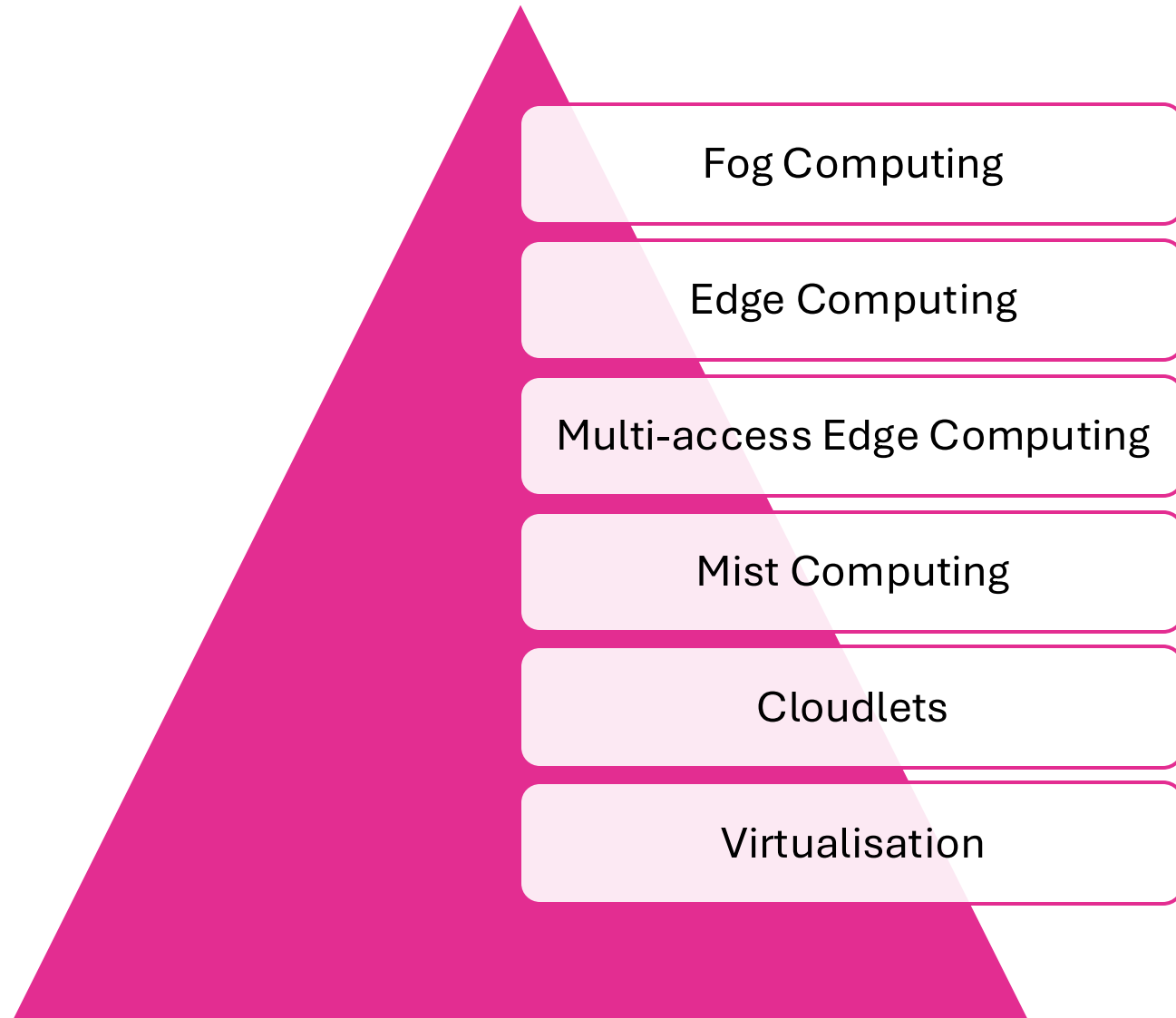
IoT applications and environments with supporting computing paradigms



Why Fog Computing for IoT

- Distributes tasks across different tiers based on complexity and resource requirements.
- Offloads simple tasks to edge devices and complex tasks to cloud or core network.
- Enables timely decision-making at the edge for critical applications like healthcare, security, and industrial automation.

Relevant Technologies



Relevant Technologies

Edge Computing

A distributed computing architecture that brings computation and data storage closer to the sources of data.

Reduced Latency: By processing data locally, edge computing significantly reduces latency, making it ideal for real-time applications like autonomous vehicles, AR/VR, and IoT

Improved Performance: Edge computing can handle large volumes of data and complex computations more efficiently than traditional cloud computing

Enhanced Security: Sensitive data can be processed and stored locally, reducing the risk of data breaches and cyberattacks.

Increased Reliability: Distributed architecture makes edge computing more resilient to failures and disruptions..

Reduced Bandwidth Consumption: Less data needs to be transmitted to the cloud, saving bandwidth and reducing costs

Relevant Technologies

Fog Computing

A specific implementation of edge computing, often focused on IoT applications

Involves a hierarchical architecture with multiple layers of computing resources.

Aims to address the limitations of cloud-centric IoT by offloading processing tasks to edge devices.

Relevant Technologies

Cloudlets

Small-scale cloud data centres located at the edge of the internet. They are designed to provide computing resources to mobile devices, reducing latency and improving the performance of resource-intensive and interactive mobile applications.

Small-scale: Cloudlets are smaller than traditional data centres, often housed in racks or containers.

Edge location: They are located closer to the end-users, typically within wireless networks or cellular base stations.

Resource-constrained: While they provide significant computing power compared to mobile devices; cloudlets have limited resources compared to large-scale data centres.

Focus on mobile devices: Cloudlets are optimized for mobile applications, providing low-latency access to computing resources

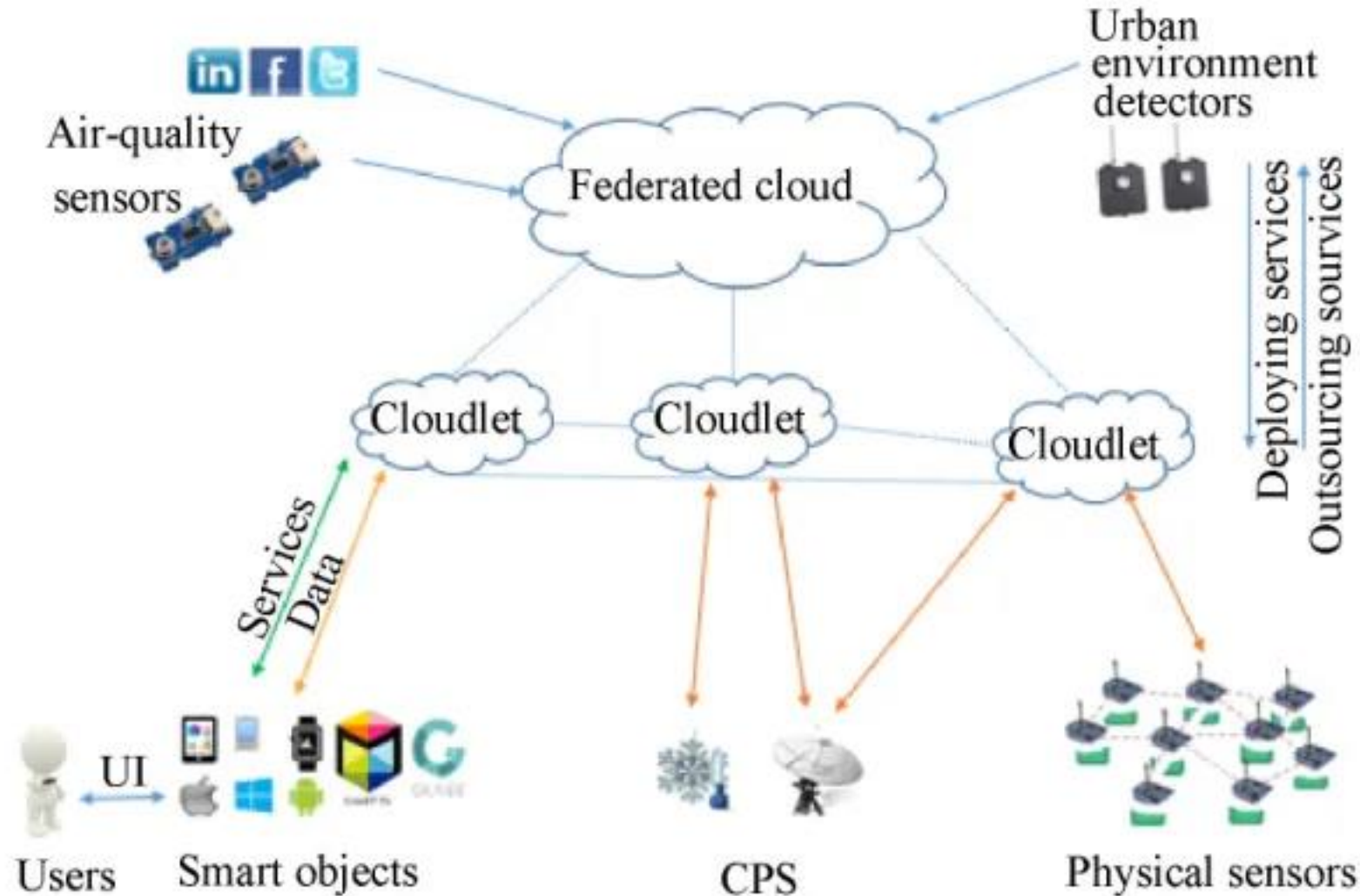
Relevant Technologies – Cloudlet Computing

- Cloudlet computing is a direction in mobile computing that shares traits with Mobile Cloud Computing (MCC) and Multi-Access Edge Computing (MEC).
- A cloudlet is a *trusted, resource-rich computer or cluster of computers with a strong internet connection*, utilized by nearby mobile devices.
- Cloudlets are *small data centers* (miniature clouds) *typically one hop away* from mobile devices.
- Cloudlet computing involves offloading computation from mobile devices to Virtual Machine (VM)-based cloudlets located on the network edge.
- Cloudlet is the middle tier of a 3-tier continuum: mobile device-cloudlet-cloud.

Relevant Technologies – Cloudlet Computing

- Cloudlet computing offers lower latency and energy consumption compared to cloud computing.
- It is intended to serve devices in the local area.
- Cloudlet computing has high potential in domains such as wearable cognitive assistance and web applications.
- Cloudlet computing differs from Mobile Access Cloud Computing (MACC) in that it requires infrastructure with virtualization and VM capability.
- Cloudlet computing supports local services for mobile clients by dividing tasks among cloudlet nodes in proximity to mobile devices.
- Fog computing offers a more generic alternative that natively supports large amounts of traffic and allows resources to be anywhere along the thing-to-cloud continuum.

Relevant Technologies – Cloudlet Computing



Relevant Technologies

Virtualization Technology that allows multiple operating systems to run on a single physical machine.

Enables efficient utilization of resources and flexible deployment of cloud services

Relevant Technologies

Multi-access

A network architecture that brings cloud computing capabilities closer to the edge of the network.

Edge

Computing (MEC)

Instead of processing data in a centralized cloud data center, processing is done at the edge of the network, like cellular base stations or Wi-Fi access points

Introduced by ETSI as a telecommunications standard.

Focuses on providing cloud-like services at the edge of the network.

Leverages existing network infrastructure like cellular base stations.

Relevant Technologies

Multi-access

MEC is an extension of mobile computing through edge computing.

Edge

Computing

(MEC)

ETSI defines MEC as a platform providing IT and cloud-computing capabilities within the **Radio Access Network (RAN)** near mobile subscribers.

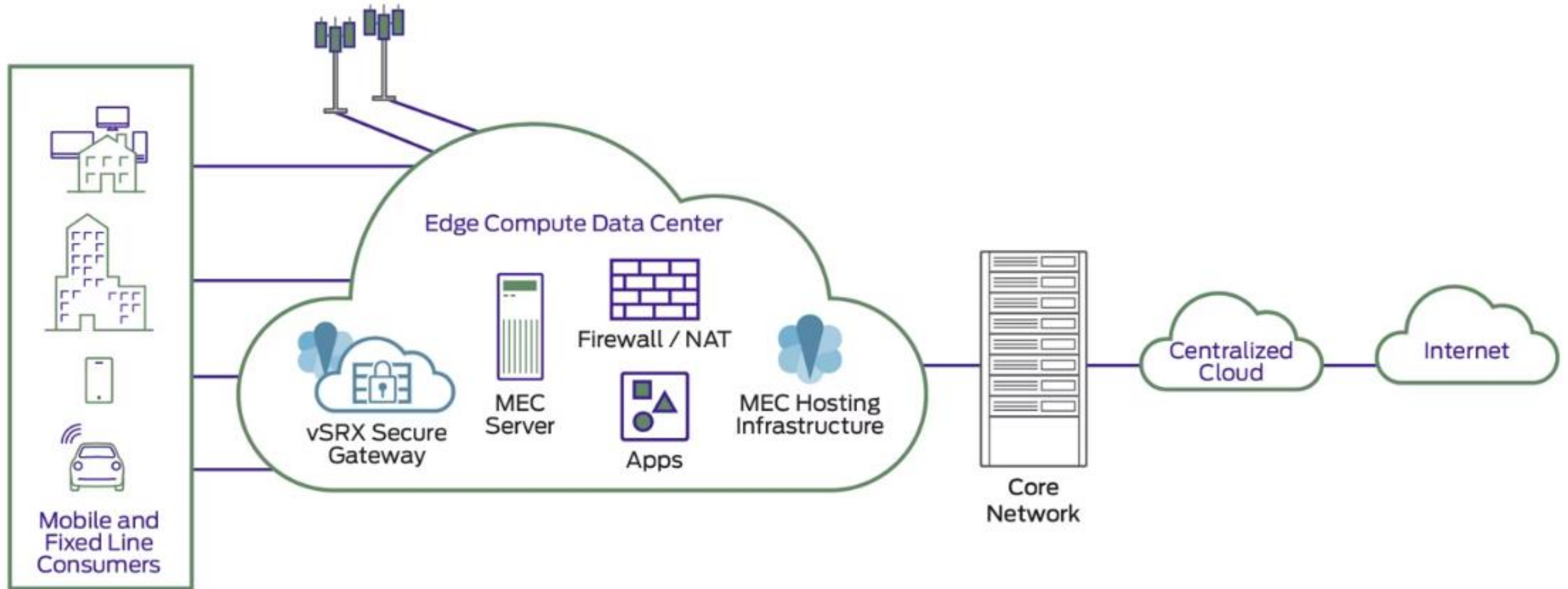
MEC extends edge computing by providing compute and storage resources near low-energy, low-resource mobile devices.

MEC operates from the edge of the Internet and can function with little to no Internet connectivity.

MEC establishes connectivity through WAN, WiFi, and cellular connections.

MEC incorporates Software-Defined Networking (SDN) and Network Function Virtualization (NFV) capabilities.

Relevant Technologies – MEC Multi Access Edge Computing



Relevant Technologies

Mist Computing

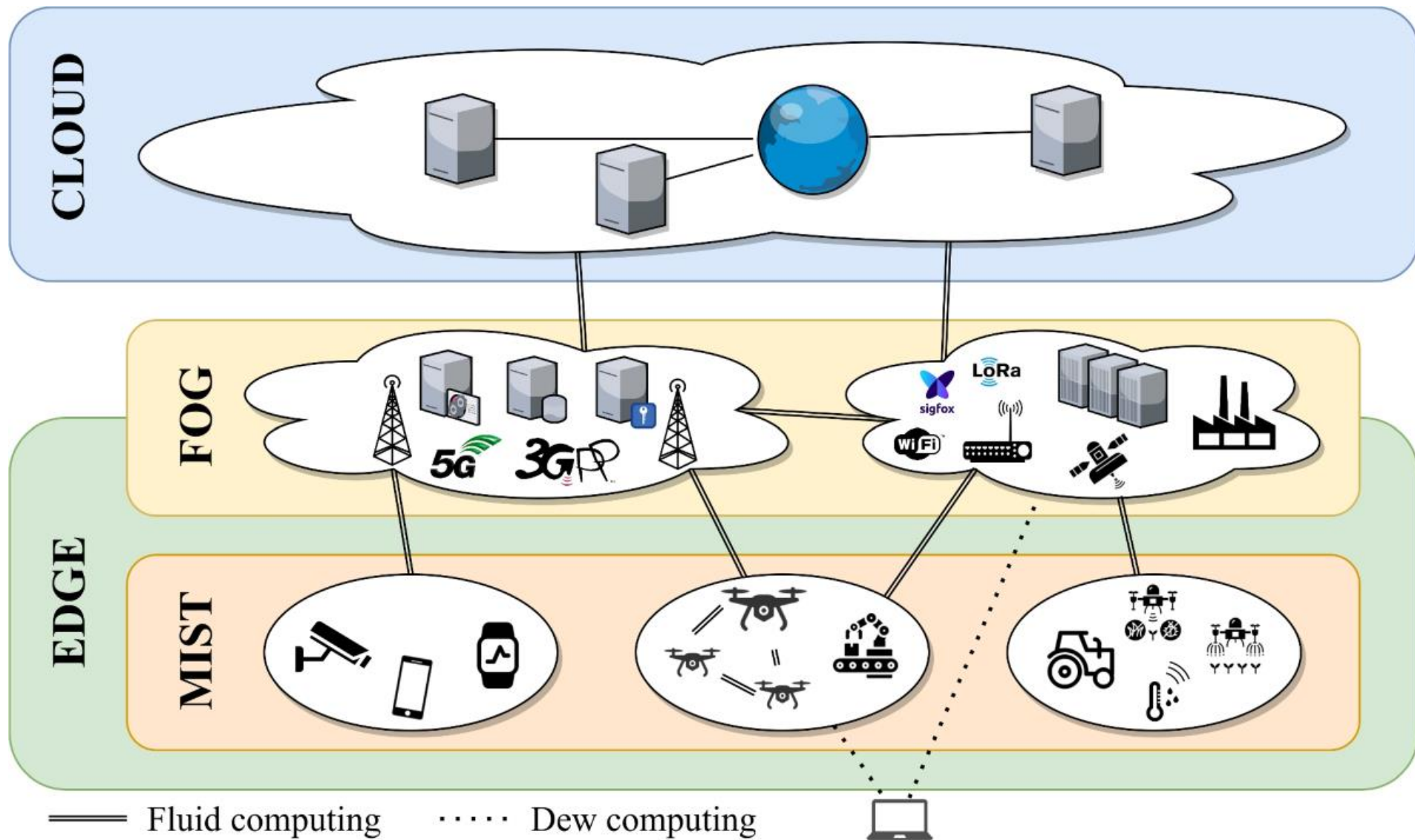
A subset of fog computing. It aims to minimize latency and enable real-time decision-making at the edge.

A specific implementation of edge computing that focuses on bringing computing capabilities to the extreme edge of the network, closer to IoT devices.

Enables IoT devices to have self-awareness and self-management capabilities.

Employs extreme edge computing to minimize latency.

Relevant Technologies – Mist Computing



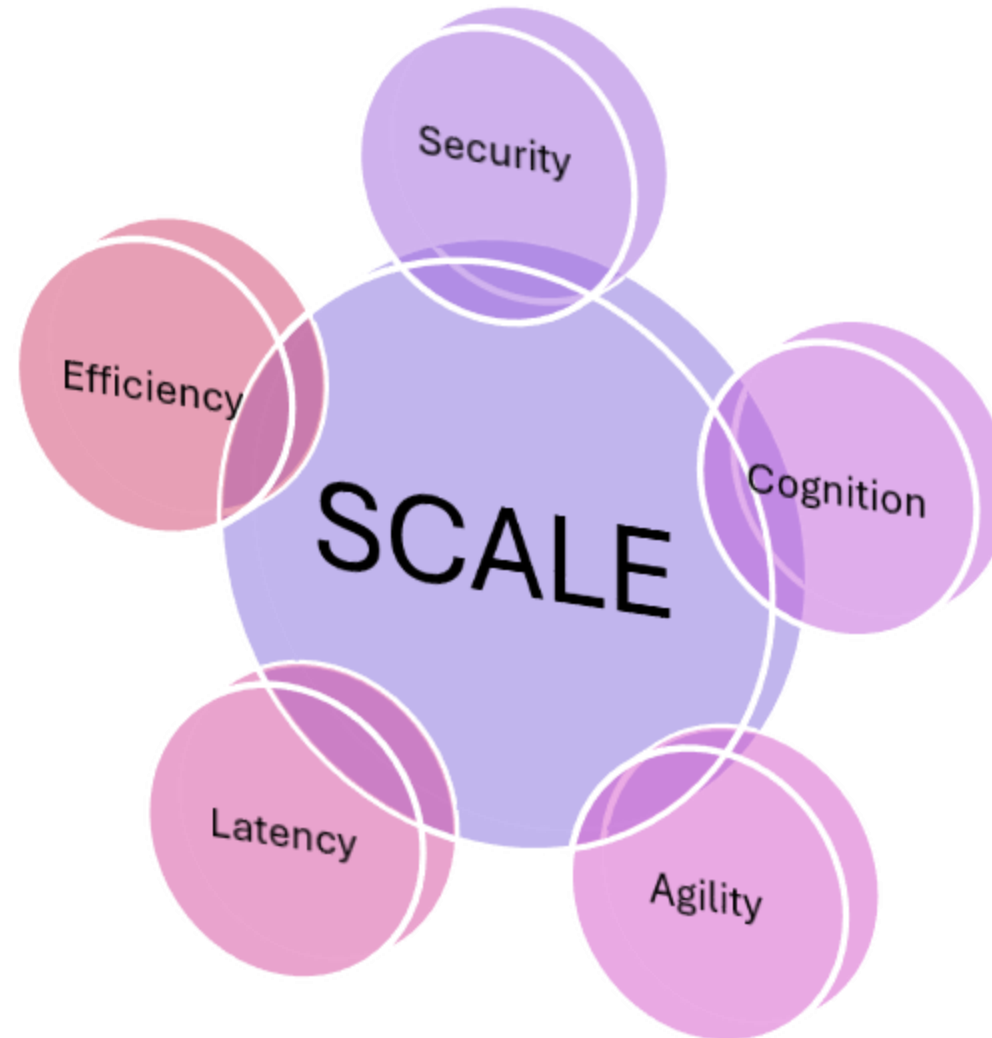
Fog Computing Vs Cloudlets

Criteria	Fog Computing	Cloudlet
Scope	Fog computing encompasses a broader range of edge computing scenarios, including IoT devices and network gateways.	Cloudlets are more focused on providing cloud services to mobile devices.
Architecture	Fog computing involves a hierarchical architecture with multiple layers of edge devices and fog nodes.	Cloudlets are typically deployed as standalone servers or virtual machines.
Goals	Fog computing aims to improve overall system performance, efficiency, and reliability.	Cloudlets primarily focus on enhancing mobile application performance and user experience.

Essence of Integration



Fog and Edge Computing Completing the Cloud



Fog and Edge Computing Completing the Cloud

Advantages of FEC - SCALE

Security

- Enables faster and more reliable security updates for IoT devices.
- Reduces vulnerability to cyberattacks by allowing for more timely security patches.

Fog and Edge Computing Completing the Cloud

Advantages of FEC - SCALE

Cognition

- Enables autonomous decision-making at the edge.
- Allows IoT devices to adapt to changing conditions and respond to events without relying on cloud-based decisions.
- Promotes self-awareness, self-organization, and self-healing capabilities in IoT devices.

Fog and Edge Computing Completing the Cloud

Advantages of FEC - SCALE

Agility

- Enables faster and more flexible deployment of IoT systems.
- Empowers individuals and small businesses to participate in providing FEC services.
- Promotes the adoption of open standards and open-source software.

Fog and Edge Computing Completing the Cloud

Advantages of FEC - SCALE

Latency:

- Reduces latency by processing data closer to the source.
- Enables real-time processing and decision-making.
- Supports time-sensitive applications like industrial automation and autonomous vehicles.

Fog and Edge Computing Completing the Cloud

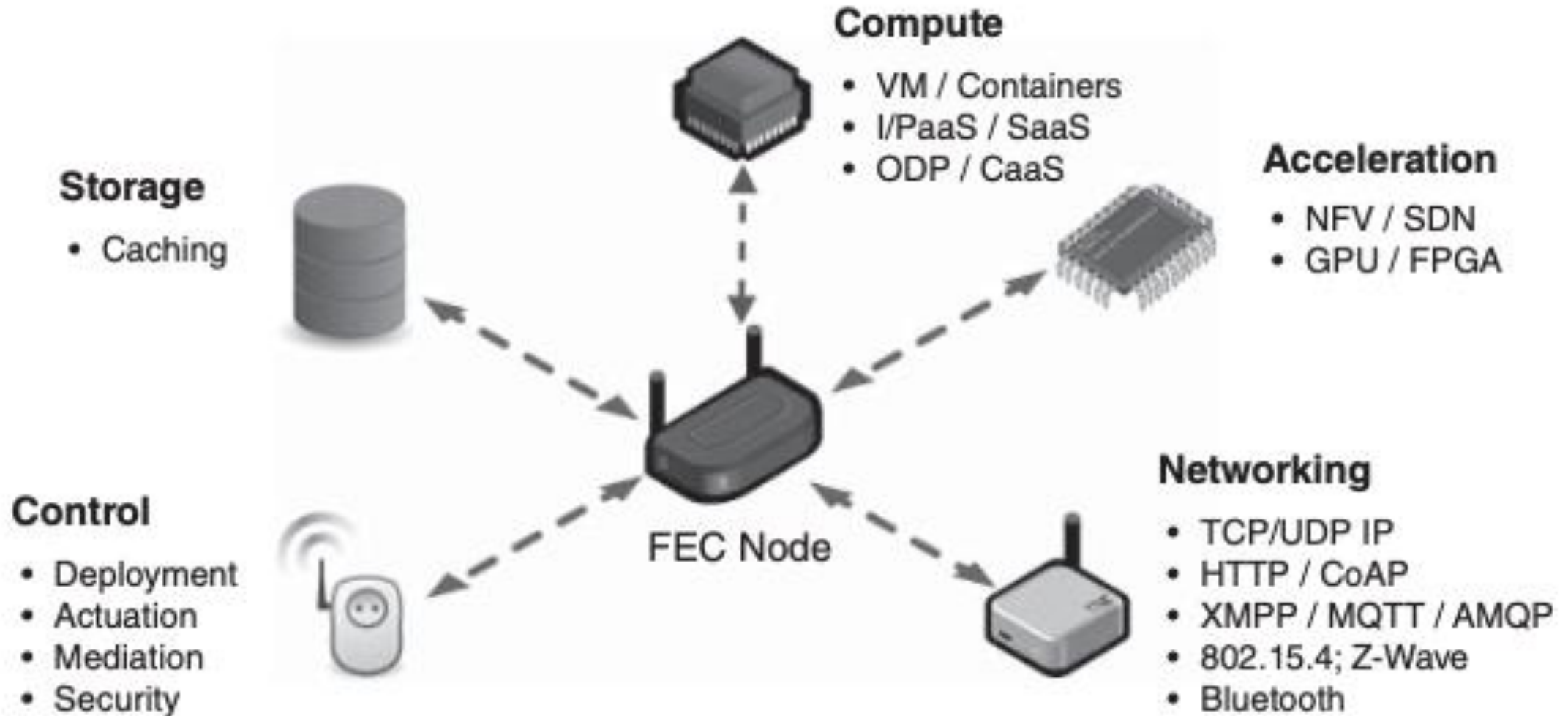
Advantages of FEC - SCALE

Efficiency:

- Improves system performance by offloading tasks to edge devices.
- Reduces bandwidth consumption and operational costs.
- Enables efficient utilization of resources.

Fog and Edge Computing Completing the Cloud

➤ How does FEC provide SCALE advantages?



Fog and Edge Computing Completing the Cloud

➤ How does FEC provide SCALE advantages?

Storage

- FEC nodes store data temporarily to improve performance.

Compute

- FEC nodes provide computing resources through I/PaaS and SaaS models.

Acceleration

- Networking Acceleration:
- Computing Acceleration:

Networking

- Vertical Networking:
- Horizontal Networking

Control

- Deployment Control
- Actuation Control
- Mediation Control
- Security Control

Fog and Edge Computing Completing the Cloud

➤ How does FEC provide SCALE advantages?

Storage: FEC nodes store data temporarily to improve performance.

Content providers can cache multimedia content closer to customers for better quality of experience.

Connected vehicles can use FEC nodes to store and share information.

Fog and Edge Computing Completing the Cloud

➤ How does FEC provide SCALE advantages?

Compute: FEC nodes provide computing resources through I/PaaS and SaaS models.

I/PaaS offers flexible platforms using VMs or CEs for customized software deployment.

SaaS provides:

On-demand data processing (ODP): Pre-installed methods for processing client data.

Context as a service (CaaS): Customized data collection, processing, and information generation.

Fog and Edge Computing Completing the Cloud

➤ How does FEC provide SCALE advantages?

Acceleration

Networking Acceleration

Traditional networks have fixed routing paths.

FEC supports network virtualization for flexible routing.

Clients can configure custom routing paths for optimal speed.

Computing Acceleration

FEC nodes can use advanced hardware like GPUs and FPGAs.

GPUs enhance complex algorithms.

FPGAs allow for reconfigurable hardware, improving efficiency and customization.

Fog and Edge Computing Completing the Cloud

➤ How does FEC provide SCALE advantages?

Networking: Vertical Networking

Connects IoT devices and cloud using IP-based protocols (TCP/UDP, HTTP, CoAP, XMPP, AMQP, MQTT).

FEC nodes act as proxies or message brokers for efficient data transfer.

Horizontal Networking

Connects IoT devices using various protocols (Bluetooth, ZigBee, Z-Wave).

IP network gateways are ideal for hosting FEC servers to bridge the gap between IoT devices and cloud.

FEC servers can control communication between IoT devices and cloud, optimizing efficiency and customization.

Fog and Edge Computing Completing the Cloud

➤ How does FEC provide SCALE advantages? - Control

Deployment Control:

- Clients can dynamically deploy and control software execution on FEC nodes.
- FEC providers can offer network topology as a service for flexible program deployment.
- Clients can control multiple FEC nodes for optimal performance.

Actuation Control:

- Cloud can delegate control of IoT devices to FEC nodes for efficient decision-making.

Mediation Control:

- FEC nodes can facilitate communication between devices from different providers.
- On-demand software updates enhance interoperability.

Security Control:

- FEC nodes ensure secure authentication, authorization, identity, and protection of the runtime environment.

Fog and Edge Computing Completing the Cloud

Deployment Control - Example

- **Automatic software updates:** Fog edge devices can automatically update software and firmware to ensure they have the latest security patches and features.
- **Container orchestration:** Fog edge devices can use container orchestration tools like Kubernetes to manage and deploy containerized applications.
- **Device provisioning:** Fog edge devices can be provisioned remotely, allowing administrators to configure and deploy devices without physical access.

Fog and Edge Computing Completing the Cloud

Actuation Control - Example

- **Real-time control of industrial equipment:** Fog edge devices can control industrial equipment in real-time, allowing for faster response times and improved efficiency.
- **Smart lighting systems:** Fog edge devices can control smart lighting systems, adjusting lighting levels based on ambient light, occupancy, and time of day.
- **Autonomous vehicles:** Fog edge devices can control autonomous vehicles, making real-time decisions about steering, acceleration, and braking.

Fog and Edge Computing Completing the Cloud

Mediation Control - Example

- **Data filtering and processing:** Fog edge devices can filter and process data in real-time, reducing the amount of data that needs to be transmitted to the cloud.
- **Protocol conversion:** Fog edge devices can convert between different protocols, allowing devices with different communication protocols to communicate with each other.
- **Data aggregation:** Fog edge devices can aggregate data from multiple sources, providing a unified view of the data

Fog and Edge Computing Completing the Cloud

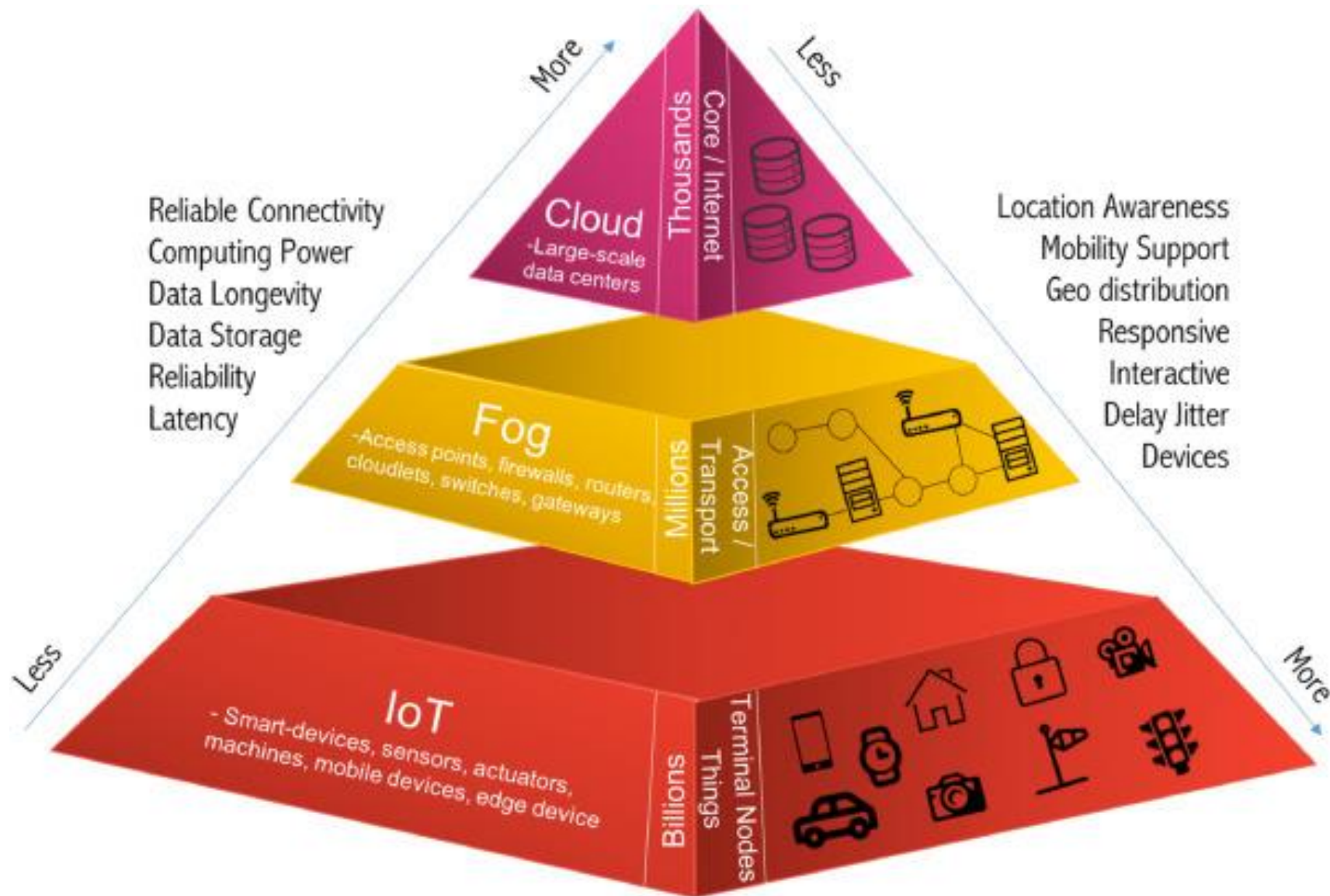
Security Control - Example

- **Intrusion detection and prevention:** Fog edge devices can detect and prevent intrusions in real-time, reducing the risk of security breaches.
- **Encryption:** Fog edge devices can encrypt data in real-time, protecting it from unauthorized access.
- **Access control:** Fog edge devices can control access to devices and data, ensuring that only authorized personnel have access.

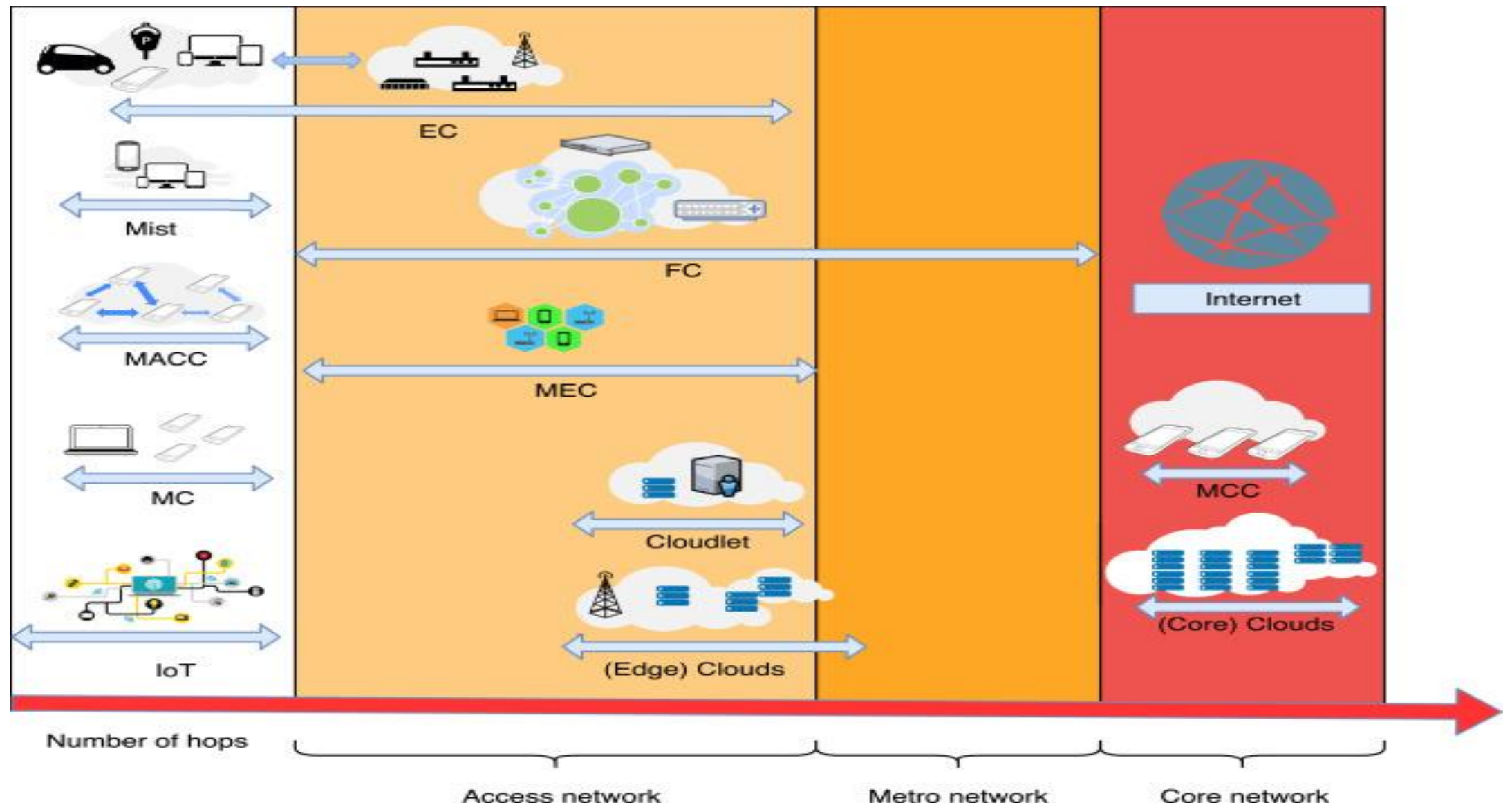
Hierarchy of Fog and Edge Computing

- CIoT systems can deploy FEC servers at three locations relative to the central cloud:
 - Inner-edge
 - Middle-edge
 - Outer-edge

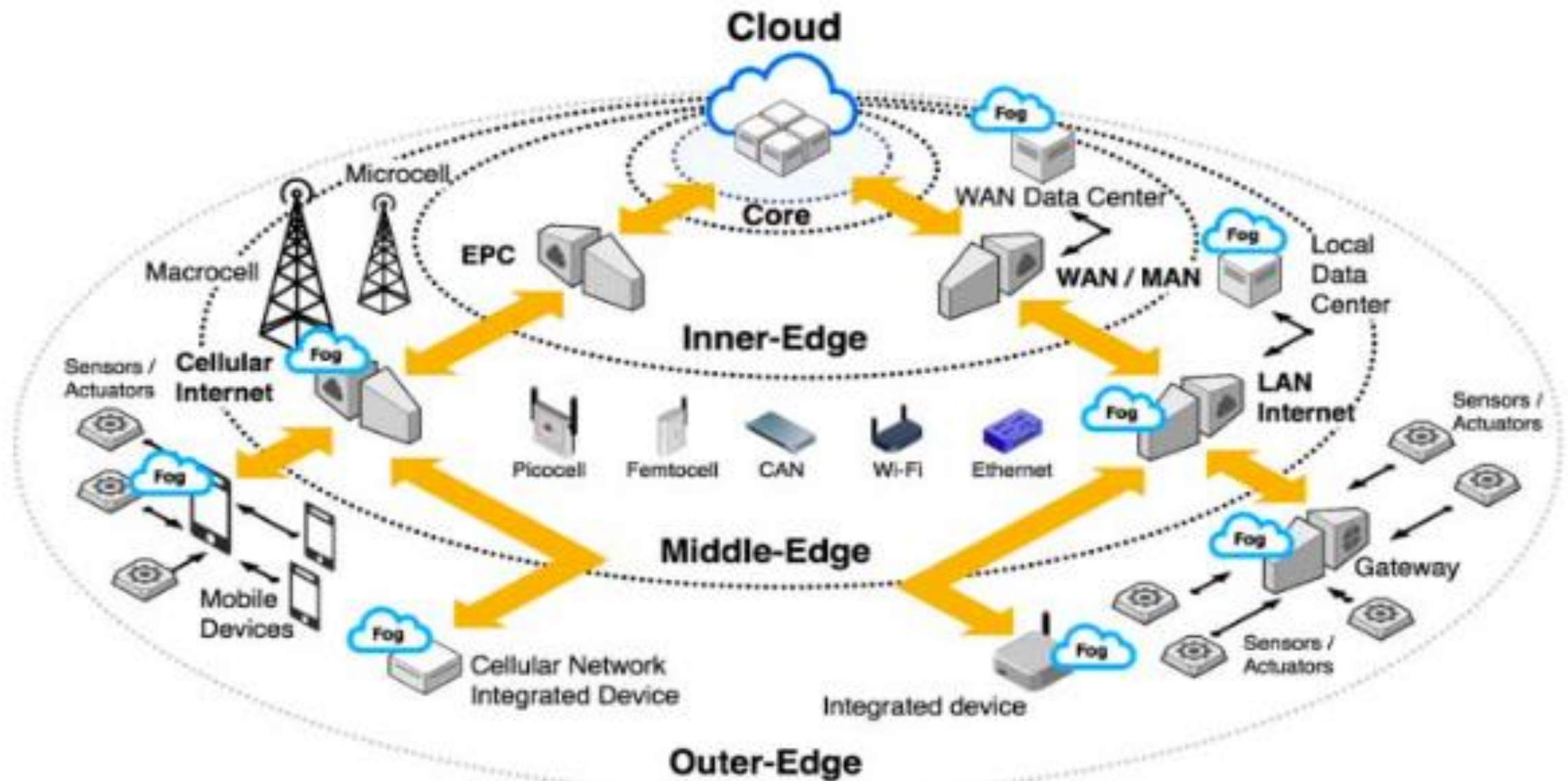
Hierarchy of Fog and Edge Computing



Hierarchy of Fog and Edge Computing



Hierarchy of Fog and Edge Computing



Hierarchy of Fog and Edge Computing

Factor	INNER EDGE	MIDDLE EDGE	OUTER EDGE
Location	Large-scale networks like WANs and MANs.	Local area networks (LANs) and cellular networks.	IoT devices & IP gateway devices.
Role:	Provides fundamental infrastructure and geo-distributed caching/processing for improved performance.	Offers computing and storage resources closer to the edge devices.	Provides computing and storage capabilities at the very edge of the network.
Work	Leverage existing data centers and network infrastructure to deploy FEC servers for enhanced web services and local cloud services.	Utilize network devices like routers and base stations as FEC hosts.	• For constraint devices: Focus on self-management and collaboration through mist computing. • For integrated devices: Explore virtualization-based FEC solutions as technology advances. • For IP gateway devices: Utilize their computing power and networking capabilities to host FEC services.

Hierarchy of Fog and Edge Computing

➤ Inner Edge

➤ Location

- Countrywide, statewide, or regional networks of enterprises, ISPs, data centers of evolved packet core (EPC), and metropolitan area networks (MAN).

➤ Initial Function

- Provides basic infrastructure to connect local networks to the global internet.

➤ Recent Trend

- Geo-distributed caching and processing for improved Quality of Experience (QoE) for web services.

Hierarchy of Fog and Edge Computing

➤ Inner Edge

➤ Examples:

- Google Edge Network collaborating with ISPs for data server distribution at ISP data centers for faster response times.
- Local cloud services offered by ISPs (AT&T, Telstra, Vodafone, Deutsche Telekom) to meet low-latency needs of local businesses.
- Fog Computing Reference: WAN-based cloud data centers can be considered as the "fog" of inner-edge.

Hierarchy of Fog and Edge Computing

➤ Middle Edge

➤ Local Area Network (LAN)

- Fog Computing Architecture: Cisco's research introduced the concept of using internet gateway devices (like Cisco IR829) to provide cloud-like services.
- Virtualization: These devices can support FEC mechanisms through virtualization technologies.
- Local Cloud/Data Center/Cloudlet: Virtualization-enabled server computers within the same subnet (one-hop range) can act as FEC nodes.

Hierarchy of Fog and Edge Computing

➤ Middle Edge

➤ Cellular Network

- Network Virtualization: Leveraging existing network virtualization technologies used in cellular networks.
- Diverse BTS Types: Macrocells, microcells, picocells, and femtocells can serve as roadside FEC hosts for various mobile IoT use cases.
- Equipment Providers: Companies like Nokia, ADLink, and Huawei are developing MEC-enabled hardware and infrastructure solutions.
- Future Trend: Widespread availability of cellular network-based FEC in various network equipment.

Hierarchy of Fog and Edge Computing

➤ Outer Edge

- Location: Front-end of the IoT network
- Device Types:
 - Constraint Devices (Sensors, Actuators)
 - Limited processing power and memory (example: Atmel ATmega328 microcontroller)
 - Field-programmable: Can be updated or reconfigured remotely.
 - Mist Computing: Enables self-management and collaboration among devices for M2M communication.

Hierarchy of Fog and Edge Computing

➤ Outer Edge

➤ Integrated Devices (Smartphones, Tablets)

- Decent processing power, networking capabilities, embedded sensors, storage
- Limited OS environment might hinder virtualization
- Promising solutions: Apache Edgent, Termux
- Future Trend: Virtualization-based FEC on integrated devices

➤ IP Gateway Devices (Hubs)

- Mediate between constrained and middle-edge devices
- Use low-energy protocols (Bluetooth Low Energy, ZigBee, Z-Wave)
- Relay communication between constrained devices and internet
- Can host virtual environments (Docker Containers Engine)
- Often used as FEC nodes in research projects

Business Models

X as a Service

Application Service

Support Service

Business Models

X as a Service (XaaS):

Covers services like infrastructure, platform, software, and networking as a service.

Allows clients to pay for hardware usage and additional service fees for software updates and technical support.

Service providers may become independent of hardware vendors, similar to companies like RedHat, IBM, or Microsoft.

The model extends to individuals or small businesses offering IndieFog solutions under the consumer-as-provider (CaP) model, e.g., MQL5 Cloud Network and Fon.

Business Models

Application Service:

Provides solutions to process data efficiently, within or outside client environments.

Examples include **Digital Twinning** for industrial optimization, **real-time traffic control** for autonomous vehicles, and **ambient assisted living (AAL)** analytics.

IndieFog providers can offer specific services, like data stream classification using platforms like Apache Edgent, for applications in local and mobile contexts.

Business Models

Support Service: Involves software management, including installation, configuration, updates, and ongoing technical support for FEC systems.

Clients may subscribe to monthly or annual plans for customized maintenance solutions.

Example: Cisco's fog computing solution with IOX-enabled equipment that supports customized operations.

Opportunities

Pre-integrated FEC runtime platforms

Allow users to easily configure and deploy applications via user-friendly interfaces.

Simplified application deployment, reduced complexity, and improved user experience.

Cisco Edge, HPE Edgeline

Open Platforms

Open platforms for FEC are in early stages with limited deployment support.

Increased flexibility, reduced vendor lock-in, and improved interoperability.

OpenStack++, WSO2-IoT Server, Apache Edgent

Opportunities

Design

Placement of FEC Servers:

Identify optimal server locations across edge tiers (inner-edge, middle-edge, outer-edge).

Resource Modelling

Use or develop abstract models to define resource requirements and interactions for FEC servers.

Implementation - Heterogeneity

Nodes at the middle and outer edge vary in hardware, protocols, and operating systems.

Vendor-specific isolated platforms (e.g., Cisco, Dell) complicate implementation, requiring custom solutions for each platform.

Time & Cost Concerns

Open platform projects exist but are not yet unified, adding to development complexity and resource investment.

Challenges

Security Risks

Pre-integrated platforms may introduce security risks if not properly secured.

Limited Customization

Pre-integrated platforms may limit customization options, making it difficult to meet specific use case requirements.

Vendor Lock-in

Pre-integrated platforms may lead to vendor lock-in, making it challenging to switch to alternative solutions.

Challenges

Complexity

Open platforms can be complex to deploy and manage, requiring specialized skills and expertise.

Limited Support

Open platforms may have limited support and documentation, making it challenging to troubleshoot issues.

Interoperability

Open platforms may face interoperability challenges, making it difficult to integrate with other systems and devices.

Device Management

Managing a large number of IoT devices can be challenging, requiring specialized device management solutions.

Opportunities & Challenges

➤ **Market Growth:**

- The market value for FEC hardware is expected to reach \$7,659 million by 2022, indicating increased availability of FEC-ready equipment like routers, switches, IP gateways, and hubs.

Opportunities

➤ **Out-of-Box Experience (OOBE): Two forms:**

➤ **OOBE-Based Equipment:**

- Pre-integrated FEC runtime platforms in devices (e.g., routers and gateways) allow users to easily configure and deploy applications via user-friendly interfaces.

➤ **OOBE-Based Software:**

- Like Microsoft Windows, enables FEC runtime environments and SCANC mechanisms on FEC-compatible devices without requiring complex configurations.

Challenges

➤ **Standardization: OOB-Based Equipment:**

- Vendors face decisions on what FEC platforms and software packages to pre-integrate.

➤ **OOB-Based Software:**

- Compatibility issues arise from heterogeneous device specifications and architectures (e.g., x86, ARM), requiring multiple software versions.

➤ **Development Costs:**

- Creating and maintaining OOB-based software is costly unless a unified standard for FEC-compatible hardware is established.

Opportunities

- Open platforms for FEC are in early stages with limited deployment support.
- A major challenge is the inflexibility of commercial devices in supporting FEC configurations, presenting opportunities for improved device designs.

Opportunities

➤ OpenStack++:

- Developed by Carnegie Mellon University.
- Features IoT-related mechanisms for connecting devices (e.g., Arduino, Raspberry Pi, Android, iOS) to the cloud via protocols like MQTT and XMPP.
- Includes WSO2-IoT Server with embedded Siddhi 3.0 for real-time streaming on embedded devices.
- Less suitable for lightweight virtualization like Docker, which is more relevant for FEC use cases.

Opportunities

➤ WSO2-IoT Server:

- Extension of the WSO2 open-source enterprise integration platform.
- Provides FEC computing capabilities to outer-edge devices by enabling real-time streaming and analytics.

Opportunities

➤ **Apache Edgent:**

- Formerly known as Quarks, contributed by IBM (edgent.apache.org).
- Supports distributed stream data processing between cloud and edge devices.
- Cloud-side integrates with major platforms like Apache Spark, Storm, and Flink.
- Edge-side supports Linux and Android OS.
- Enables dynamic migration of stream data processing between cloud and edge for edge analytics.

Challenges

- Current platforms lack end-to-end support across all layers of edge networks.
- Commercial devices need enhancement to support FEC configurations effectively.
- The shift toward containerization (e.g., Docker) indicates a need for platforms optimized for lightweight virtualization in FEC deployments.

Opportunities – Life Cycle Phases

Design:

- Placement of FEC Servers:
 - Identify optimal server locations across edge tiers (inner-edge, middle-edge, outer-edge).
- Resource Modeling:
 - Use or develop abstract models to define resource requirements and interactions for FEC servers.

Opportunities – Life Cycle Phases

➤ **Implementation:**

➤ **Heterogeneity Challenges:**

- Nodes at the middle and outer edge vary in hardware, protocols, and operating systems.
- Vendor-specific isolated platforms (e.g., Cisco, Dell) complicate implementation, requiring custom solutions for each platform.

➤ **Time and Cost Concerns:**

- Open platform projects exist but are not yet unified, adding to development complexity and resource investment.

Opportunities – Life Cycle Phases

➤ Adjustment:

➤ Runtime Adaptability:

- Systems must dynamically schedule, deploy, and terminate runtime environments (e.g., VMs, containers) to optimize performance.

➤ Dynamic Migration:

- Runtime environments and application methods should move between FEC nodes based on context, raising challenges in:
 - **Reliability:**
 - Ensuring successful migration with minimal failures.
 - **Latency:**
 - Reducing delays caused by wireless and mobile communication interruptions, particularly at the far-edge.

Challenges – Life Cycle Phases

- Supporting reliable software migration across FEC nodes.
- Minimizing latency during runtime adjustments, especially in dynamic outdoor environments.
- Addressing signal interruptions in wireless communication for code deployment and reconfiguration.